



Finding logic expression from the truth table

There are two methods to find the logic expression from the truth table:

Sum of products (SOP)

Product of the sum (POS)

Sum of products: In this method we select the rows with the output of 1, and find the corresponding logical combinations at the input.

For example

consider this truth table,

1. Identify the rows

with output of 1.

(rows: 0, 1, 4, 6)

row	A	B	C	D (output)
0	0	0	0	1
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	1
7	1	1	1	0

2. write a logical product for each of these rows.

row 0 : $\overline{A} \overline{B} \overline{C}$

row 1 : $\overline{A} \overline{B} C$

row 4 : $A \overline{B} \overline{C}$

row 6 : $A B \overline{C}$

These are called minterms.

$$D = \sum m(0, 1, 4, 6)$$

↑
minterm

3. write an expression for the output variable as logical sum of the minterms.

$$D = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + AB\bar{C}$$

Product of the sum: In this method we select the rows with the output of 0, which gives the expression for the inverted output, and find the corresponding logical combinations for that. Then use DeMorgan's law to invert it and find the expression for the output.

For the table in the

previous example:

1. Identify the rows with the output value of zero.

(rows: 2, 3, 5, 7).

Row	A	B	C	D (output)
0	0	0	0	1
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	1
7	1	1	1	0

2. write a logical sum for each of these rows.

row 2: $A + \bar{B} + C$

row 3: $A + \bar{B} + \bar{C}$

row 5: $\bar{A} + B + \bar{C}$

row 7: $\bar{A} + \bar{B} + \bar{C}$

These are called maxterms

$$D = \prod M(2, 3, 5, 7)$$

↑ maxterm
↑ product

3. write an expression for the output as logical product of the maxterms.

$$D = (A + \bar{B} + C) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{C})$$

This expression can be directly derived from the logical products of inputs for the rows that make the output D zero. Therefore, they give the expression for $\bar{D}$.

$$\bar{D} = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

using De Morgan's Law:

$$\Rightarrow D = \bar{\bar{D}} = \overline{\bar{A}\bar{B}\bar{C}} \cdot \overline{\bar{A}B\bar{C}} \cdot \overline{A\bar{B}\bar{C}} \cdot \overline{ABC}$$

$$D = (A + \bar{B} + C) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + B + \bar{C}) \cdot (\bar{A} + \bar{B} + \bar{C})$$

Karnaugh Map

A complicated logic expression can be simplified using Boolean algebra. However, it may not lead to the simplest form of a logic expression.

A systematic method for minimization of a logic expression is Karnaugh map, which is a graphical approach. It is an array of squares. Each square, represents a row in the truth table.

In karnaugh map, in each pair of adjacent squares only one variable is different.

A \ B	0	1
0	$\bar{A}\bar{B}$	$\bar{A}B$
1	$A\bar{B}$	AB

2-Variable map

A \ BC	00	01	11	10
0	$\bar{A}\bar{B}\bar{C}$	$\bar{A}\bar{B}C$	$\bar{A}B\bar{C}$	$\bar{A}BC$
1	$A\bar{B}\bar{C}$	$A\bar{B}C$	$AB\bar{C}$	ABC

3-Variable map

AB \ CD	00	01	11	10
00	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$
01	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BC\bar{D}$	$\bar{A}BCD$
11	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABC\bar{D}$	$ABCD$
10	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}C\bar{D}$	$\bar{A}\bar{B}CD$

4-Variable map

K-map for $\bar{A}B + AB$:

A \ B	0	1
0		1
1		1

K-map for $\bar{A}B\bar{C} + A\bar{B}C$:

A \ BC	00	01	11	10
0			1	
1		1		

Finding minimum SOP

Create a K-map from the desired output.

Look for the least number of groups of 2, 4, or 8 that covers the 1's. (Try to find the largest group).

Find the minimum SOP by the logical sum of the expression of each group.

Ex. Find the minimum SOP for F using K-map.

$$F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}\bar{C}D + A\bar{B}C\bar{D}$$

$$F = \bar{A}D + \bar{A}\bar{B}C \quad (\text{minimum SOP})$$

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11				
10			1	1

$\bar{A}D$ (grouping the top two rows)
 $\bar{A}\bar{B}C$ (grouping the bottom-right two cells)

Ex. A logic circuit has inputs A, B, C, D.

The output is given by $E = \sum m(0, 3, 7, 8, 11, 15)$.

Find the minimum SOP for E.

minimum SOP:

$$E = CD + \bar{B}\bar{C}\bar{D}$$

CD \ AB	00	01	11	10
00	1 0	1	1 3	2
01	4	5	1 7	6
11	12	13	1 15	14
10	1 8	9	1 11	10

$\bar{B}\bar{C}\bar{D}$ CD

Finding minimum POS

Create a K-map with the inverted output (replace output of 1 by 0 and vice versa).

Look for the least number of groups of 2, 4, or 8 that covers the 1's. (Try to find the largest group).

Find the minimum SOP by the logical sum of the expression of each group.

Use DeMorgan's law to convert SOP to POS.

Ex. In the previous example, find the minimum POS for the output E.

		CD			
		00	01	11	10
AB	00	0	1	0	1
	01	1	1	0	1
	11	1	1	0	1
	10	0	1	0	1

$B\bar{C}$ → (points to row 11)
 $\bar{C}D$ → (points to column 01)
 $C\bar{D}$ → (points to column 10)

k-map for $\bar{E}$

invert
←

		CD			
		00	01	11	10
AB	00	1		1	
	01			1	
	11			1	
	10	1		1	

k-map for E
 (original k-map)

$$\bar{E} = C\bar{D} + \bar{C}D + B\bar{C}$$

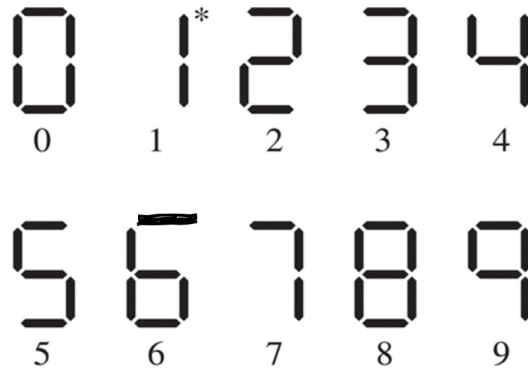
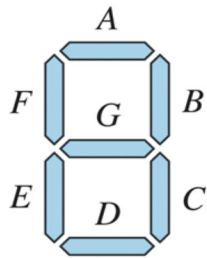
$$E = \overline{C\bar{D} + \bar{C}D + B\bar{C}} = \overline{C\bar{D}} \cdot \overline{\bar{C}D} \cdot \overline{B\bar{C}}$$

$$E = (\bar{C} + D) \cdot (C + \bar{D}) \cdot (\bar{B} + C)$$

BCD to 7-segment display

The 7-segment display is used to display numbers 0 to 9, using 7 independent LEDs. The input is in BCD (Binary Coded Decimal) and the output is the display of the equivalent input decimal number. For example if the input is 0011=3, the segments A, B, C, D and G are on to display 3.

Find the logic expression for control of segment 'A'.



(a) Segment designations

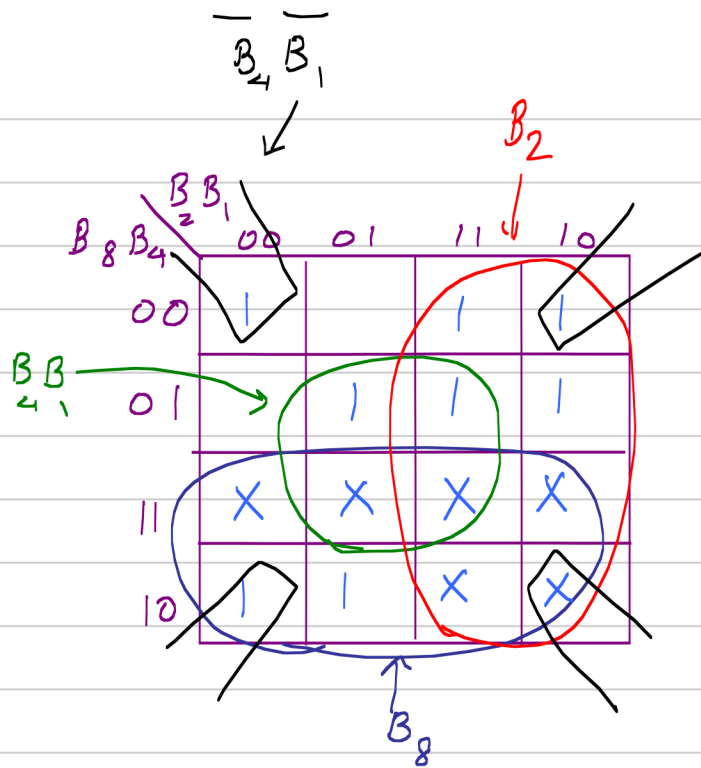
(b) Displays for digits 0 through 9

*Segments B and C are on

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	B_8	B_4	B_2	B_1	A
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
	1	0	1	0	X
	1	0	1	1	X
	1	1	0	0	X
	1	1	0	1	X
	1	1	1	0	X
	1	1	1	1	X

"don't care" entries of the truth table can be considered "0" or "1", whichever leads to the minimum expression.



$$A = B_2 + B_8 + B_4 B_1 + \overline{B_4} \overline{B_1}$$

